

# Certified Analytical Bounds for Multi-Phase Hypersonic Reachability

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**Abstract**

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- 1.1 The computational limitations of grid-based Hamilton-Jacobi-Isaacs evaluations**
- 1.2 Thesis statement**
- 1.3 Outline of contributions**

## 2 The Model Problem

- 2.1 Abstract formulation of the multi-phase optimal control problem
- 2.2 The degenerate case: equilibrium glide
- 2.3 The oscillatory and hybrid cases: FOBS and Sänger-type skips
- 2.4 Quantity of interest and uncertainty

### 3 Infinite-Dimensional Embedding and Measure Theory

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### 5.1 Filippov-Ważewski relaxation

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## 6 Verification and Results

### 6.1 Validating the degenerate limit

### 6.2 The oscillatory and synergetic demonstration

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## 7 Discussion

### 7.1 System diagnostics and computational cost

### 7.2 Implications for defense modeling

## References

- A Point-Mass Equations of Motion**
- B The Extremal Control Law**
- C Sum-of-Squares Programming and Formulation**
- D Stochastic Extension**